Problem 25.4

There is \$40,000 in a fund which is accumulating at 4% per annum convertible continuously. If money is withdrawn continuously at the rate of \$2,400 per annum, how long will the fund last?

Solution

The fund earns interest at a constant force of interest

$$\delta = 0.04.$$

The equivalent annual effective interest rate i satisfies

$$1 + i = e^{\delta}.$$

Thus,

$$i = e^{0.04} - 1 = 0.0408108.$$

The withdrawals form a continuous annuity at a rate of 2,400 per year. If the fund lasts for n years, then the present value of the withdrawals equals the initial fund:

$$40000 = 2400\bar{a}_{\overline{n}|}.$$

For a continuous annuity,

$$\bar{a}_{\overline{n}|} = \frac{1 - v^n}{\delta},$$

where $v = (1+i)^{-1} = e^{-\delta}$. Therefore,

$$40000 = 2400\left(\frac{1 - v^n}{0.04}\right).$$

Divide by 2400:

$$\frac{40000}{2400} = \frac{1 - v^n}{0.04}.$$

Since $40000/2400 = 50/3$,

$$\frac{50}{3} = \frac{1 - v^n}{0.04}.$$

Thus,

$$\frac{2}{3} = 1 - v^n.$$

So

$$v^n = \frac{1}{3}.$$

Because $v = (1+i)^{-1}$,

$$(1+i)^{-n} = \frac{1}{3}.$$

Therefore,

$$(1+i)^n = 3.$$

Taking natural logarithms,

$$n \ln(1+i) = \ln 3.$$

Since $\ln(1+i) = \delta = 0.04$,

$$n = \frac{\ln 3}{0.04} = 27.465307.$$

Thus, the fund will last approximately

$$\boxed{27.47 \text{ years}}.$$